

Imagine a leaf falls inside of an outdoor hot-tub. Once the leaf falls beneath the surface of the water, the jets of the hot tub will push the leaf around. Specifically, at any given point in time, the hot tub's jets are applying a force to any objects in the water—one can think of this influence as applying a velocity-vector at that location in the hot tub.

If you were to 'zoom out' and look at the entire hot-tub, you can imagine a collection of these various vectors – this is called a **vector-field**. In this manner, you can 'see' the leaf move around this vector field and get pushed around by the vectors. A 3-space vector field is traditionally given by

$$\mathbf{F}(x, y, z) = \langle M(x, y, z), N(x, y, z), P(x, y, z) \rangle.$$

Continuity and differentiability of $\mathbf{F}$ is inherited by its components.

Example. The unit tangent vectors $\mathbf{T}$ and principle unit normal vectors $\mathbf{N}$ of a vector function $\mathbf{r}$, as well as the gradient vectors ∇f of a surface f all form vector-fields:

Recall that a vector-valued function is something like $\mathbf{r}(t) = \langle g(t), h(t), k(t) \rangle$. Thus, vector-fields $\mathbf{F}(x, y, z) = \langle M(x, y, z), N(x, y, z), P(x, y, z) \rangle$ are just multi-variable versions of vector-functions. We've learned how to compute line integrals of functions f defined across curves C traced out by vector-valued functions. So how do we do for their multi-variable counterparts?

Imagine a particle moving through the vector-field $\mathbf{F}$:

Along the way, the vector-field's vectors will help 'push' the particle along – but the only part that matters is the part pointing in the direction of movement. That is, in the direction of $\mathbf{T}$! To find this 'push' contribution, we project the vector-field's vector $\mathbf{F}$ onto the unit tangent vector $\mathbf{T}$. This scalar-projection is exactly

$$f(x, y, z) := \text{scalar-projection of } \mathbf{F} \text{ onto } \mathbf{T} = \frac{\mathbf{F} \cdot \mathbf{T}}{\|\mathbf{T}\|} = \mathbf{F} \cdot \mathbf{T} = \mathbf{F} \cdot \frac{d\mathbf{r}}{ds}$$

Therefore, *the line integral of $\mathbf{F}$ along C* is

$$\int_C f(x, y, z) ds = \int_C \left(\mathbf{F} \cdot \frac{d\mathbf{r}}{ds} \right) ds = \int_C \mathbf{F} \cdot d\mathbf{r}.$$

Thus, since $\mathbf{T}(t) = \frac{\mathbf{v}(t)}{\|\mathbf{v}(t)\|} = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}$ and $ds = \|\mathbf{v}(t)\|dt = \|\mathbf{r}'(t)\|dt$, we often compute this instead:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C f(x, y, z) ds = \int_C \mathbf{F} \cdot \mathbf{T} ds = \int_a^b \left(\mathbf{F}(\mathbf{r}(t)) \cdot \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|} \right) \|\mathbf{r}'(t)\| dt = \int_a^b \left(\mathbf{F}(\mathbf{r}(t)) \cdot \frac{d\mathbf{r}}{dt} \right) dt,$$

where C is traced out from $t = a$ to $t = b$.

Example. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F}(x, y, z) = \langle z, xy, -y^2 \rangle$ along the curve C given by $\mathbf{r}(t) = \langle t^2, t, \sqrt{t} \rangle$ for $0 \leq t \leq 1$.

Revisiting the hot-tub, suppose the vector-field $\mathbf{F}$ represents a velocity-field of the water flow. In this case, the line-integral of $\mathbf{F}$ across a curve C in the water is tracking the fluid's *flow* along C . If C is a closed curve, then the line-integral is tracking the fluid's *circulation*. Notationally,

$$\text{Flow} = \int_C \mathbf{F} \cdot \mathbf{T} ds \quad \text{and} \quad \text{Circulation} = \oint_C \mathbf{F} \cdot \mathbf{T} ds.$$

If C is in the xy -plane traveling counter-clockwise, then we can further deduce that

$$\mathbf{F} \cdot \mathbf{T} ds = \mathbf{F} \cdot \frac{d\mathbf{r}}{ds} ds = \langle M(x, y), N(x, y) \rangle \cdot \left\langle \frac{dx}{ds}, \frac{dy}{ds} \right\rangle ds = \left(M \frac{dx}{ds} + N \frac{dy}{ds} \right) ds = M dx + N dy.$$

Example. Consider the field $\mathbf{F}(x, y) = \langle x - y, x \rangle$. Draw this vector-field on the xy -plane and then find the circulation of the $\mathbf{F}$ around the circle $\mathbf{r}(t) = \langle \cos(t), \sin(t) \rangle$ for $0 \leq t \leq 2\pi$.

A curve C is *simple* if it doesn't cross itself. The 'standard orientation' for a curve C is counter-clockwise. Imagine again a 2-space curve C traced out in the xy -plane:

If flow measures how an ambient fluid influences an object moving *along* C , we can ask the 'orthogonal question' and inquire about the rate at which the fluid is entering or leaving the region enclosed by C —this is called **flux**. If $\mathbf{n}$ is the outward-pointing unit normal vector on C , then

$$\text{Flux of } \mathbf{F} \text{ across } C = \oint_C \mathbf{F} \cdot \mathbf{n} \, ds.$$

Positive $\mathbf{n}$ measures how fast the fluid is *leaving* the region, and $-\mathbf{n}$ measures how fast the fluid is *entering* the region.¹

So how do we find $\mathbf{n}$? If C is in the xy -plane traveling counter-clockwise, then there is a simple answer: $\mathbf{n} = \mathbf{T} \times \mathbf{k}$, where $\mathbf{k} = \langle 0, 0, 1 \rangle$. With this choice, the above formula for flux becomes:

Theorem. If $\mathbf{F}(x, y) = \langle M(x, y), N(x, y) \rangle$ is a vector-field and C is a simple, closed curve in the xy -plane, then

$$\text{Flux of } \mathbf{F} \text{ across } C = \oint_C \mathbf{F} \cdot \mathbf{n} \, ds = \oint_C M(x, y) \, dy - N(x, y) \, dx,$$

where we can evaluate this integral using any smooth parametrization of C that traces itself out exactly once in the counter-clockwise direction.

Example. Find the flux of $\mathbf{F}(x, y) = \langle x - y, x \rangle$ across the unit circle in the xy -plane.

¹Note: We can still compute flux over non-closed curves by using a line integral instead of a contour integral.